

Jose Bande Serrano

Full-stack Software Engineer

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Professional Summary

Full-stack Software Engineer with 10+ years of experience delivering production full-stack systems with hands-on ownership across backend services, APIs, data workflows, and modern frontend applications. Offers strong evidence in Python, Django, FastAPI, PostgreSQL, Redis, and REST APIs, with additional experience across Codex, OpenAI, AI Agents, and AI-Assisted Development. Background includes platform architecture, CI/CD, cloud delivery, and building scalable workflow-driven applications.

Selected Skills

Python, Django, FastAPI, PostgreSQL, Redis, REST APIs, React, Next.js, TypeScript, CI/CD, Docker, System Architecture, Node.js, Azure, Project Management, Team Leadership

Professional Experience

NBCUniversal - Full-Stack Engineer, Compliance Automation

2024-09 - present | New York, NY

Built and delivered a production compliance automation platform combining a high-performance Next.js SPA, backend workflows, ETLs, and reporting systems.

- Built a React.js and Next.js application that is in production and used by hundreds of NBCUniversal employees.
- Implemented AI bots integrated with Microsoft Teams.
- Developed a React application embedded inside a model-driven Power Apps application.
- Designed and implemented tens of Dataverse tables to support the application's data model.
- Created dozens of Power Automate flows and dataflows to support compliance workflows and data movement.

Enerex - AI Development Consultor

2023-11 - present | Remote - New Jersey

Architected AI-powered automation platforms spanning user-facing applications, asynchronous workflow orchestration, and cloud delivery for contract-processing systems.

- Designed the platform around three core services: a connector API for task intake and third-party email ingestion, a monitor application for task and flow persistence plus observability, and a queue-triggered Azure Functions worker tier for agent execution.
- Architected the "Unattended Mode" platform, an asynchronous work-farm architecture for AI-driven email and contract processing without requiring users to wait in front of a UI.
- Built the Outlook add-in frontend in React and Next.js, delivering a complex UI for reviewing, correcting, and submitting extracted contract data.
- Built an AI-powered Outlook add-in to automate contract processing workflows for energy retail market users.
- Built production applications with Codex as a primary development tool, accelerating implementation across frontend, AI service, and delivery work.

Broadcasting Platforms - Senior Frontend Engineer & Team Lead

2019-11 - 2022-12 | Remote

Led and built the core user-facing frontend applications of a real-time streaming platform, with deep ownership across Vue.js architecture, design-system implementation, and WebSocket-driven interaction systems.

- Improved frontend delivery performance by introducing HTTP/2, asset compression, and image conversion to WebP format.
- Built two core user-facing streaming applications in Vue.js that formed the frontend backbone of the platform.
- Designed and implemented an SEO-focused rendering solution that served server-side-rendered content to web crawlers while preserving the SPA experience for regular users.
- Implemented demanding frontend experiences from professional Figma design specifications with strong attention to design fidelity and usability.
- Used Storybook as a design-system layer to centralize, standardize, and control component styling across the platform.

First Due - Senior Software Engineer

2017-07 - 2019-11 | Remote

Built large-scale web crawlers and ETL pipelines with Python and event-driven backend systems.

- Designed and maintained PostgreSQL databases supporting large-scale data collection workloads.
- Implemented tens of Python web crawlers with the Scrapy framework to extract critical company data from public websites.
- Built crawler orchestration and monitoring capabilities using Django, FastAPI, and RabbitMQ.
- Improved control and accountability of company web-crawling operations through centralized management and visibility.
- Worked in Linux environments while building and operating Python-based data extraction and ETL systems.

Liberty Power - .NET/SQL Consultant

2016-05 - 2017-06 | Remote

Modernized legacy .NET applications and improved database quality and performance with SQL work.

- Implemented and ran SQL queries to investigate, validate, and repair data issues.
- Implemented numerous stored procedures to support application behavior and database operations.
- Migrated projects from Visual Basic to ASP.NET as part of legacy application modernization work.
- Performed extensive database data consistency reviews and corrections across business-critical datasets.

Projects

AI Assisted Invoice Processor

An application to help finance teams collect invoices, extract structured data, review exceptions, and move approved records into downstream reporting without juggling disconnected inboxes, sheets, and shared folders.

- Built the frontend with React.js, Next.js, Tailwind CSS, Lucide React, and TanStack to support real-time, data-driven invoice operations.
- Built the backend with Node.js, Express.js, and Prisma over SQL Server hosted in Azure.
- Deployed the system in Azure with CI/CD through Azure DevOps.
- Integrated Nylas and Google services to enable controlled access to user email inboxes for invoice ingestion.

Education

Bachelor's Degree, Engineer in Electronics

Ciudad Universitaria Jose A. Echeverria (CUJAE) | Havana, Cuba | 2004 - 2009